

In the previous lecture, we defined the generalized eigenspace of an eigenvalue λ for a linear operator $T : V \rightarrow V$. Clearly, $E_\lambda \subseteq G_\lambda$. In fact,

$$G_\lambda = \bigcup_{k \geq 1} \ker((T - \lambda I)^k).$$

We know by the Stabilization Theorem that there is an integer $m \leq n$ for which

$$\ker((T - \lambda I)^m) = \dots = \ker((T - \lambda I)^n) = \dots$$

Hence

$$G_\lambda = \ker((T - \lambda I)^n)$$

and therefore, G_λ is a subspace of V .

Proposition

Let T be a linear operator on a finite-dimensional vector space V . Suppose $\lambda_1, \lambda_2, \dots, \lambda_k$ are distinct eigenvalues of T . Let v_i be a generalized eigenvector corresponding to λ_i for each $i = 1, \dots, k$. Then $\{v_1, v_2, \dots, v_k\}$ is a linearly independent set.

Proof: Suppose we have a trivial linear combination of v_i 's:

$$c_1 v_1 + \dots + c_k v_k = \mathbf{0}$$

We show $a_1 = 0$. A similar method will show $a_i = 0$ for any i .

By definition, $(T - \lambda_i I)^n(v_i) = \mathbf{0}$. Now consider $P(T) : V \rightarrow V$ defined as

$$P(T) = (T - \lambda_2 I)^n (T - \lambda_3 I)^n \dots (T - \lambda_k I)^n.$$

Applying $P(T)$ to both sides, we have

$$P(T)(a_1 v_1) + P(T)(a_2 v_2) + \dots + P(T)(a_n v_n) = \mathbf{0}.$$

Note that the terms of $P(T)$ commute. For any $1 \leq \alpha, \beta \leq k$,

$$(T - \lambda_\alpha I)(T - \lambda_\beta I) = T^2 - \lambda_\alpha T - \lambda_\beta T + \lambda_\alpha \lambda_\beta I = (T - \lambda_\beta I)(T - \lambda_\alpha I).$$

Hence,

$$(T - \lambda_\alpha I)^n (T - \lambda_\beta I)^n = (T - \lambda_\beta I)^n (T - \lambda_\alpha I)^n.$$

This means that, given $i \geq 2$,

$$\begin{aligned} P(T)(a_i v_i) &= (T - \lambda_1 I)^n (T - \lambda_2 I)^n \dots (T - \lambda_k I)^n (a_i v_i) \\ &= (T - \lambda_1 I)^n \dots (T - \lambda_i I)^n (a_i v_i) = \mathbf{0}. \end{aligned}$$

It follows that $a_1 P(T)(v_1) = \mathbf{0}$, so $a_1 = 0$. ■

Proposition

Let T be a linear operator on a complex finite-dimensional vector space V , and let λ be an eigenvalue of T . Then G_λ is T -invariant.

Proof: Let $v \in G_\lambda$. So $(T - \lambda I)^n(v) = \mathbf{0}$. Then

$$\begin{aligned}(T - \lambda I)^n(T(v)) &= (T - \lambda I)^n(T - 0I)(v) \\ &= (T - 0I)(T - \lambda I)^n(v) = \mathbf{0}.\end{aligned}$$

Hence $T(v) \in G_\lambda$. ■